DAY-3 – CONFIGURING BASIC NETWORK HARDWARE

Course: Diploma in Cyber Security

Topic: IP Addressing Basics and Network Configuration

Name: HET ACHARYA

Theory – 30 min

IP addressing basics and network configuration

Practical – 30 min

Set up a router with default settings

TASK 1 – ROUTER ADMIN PANEL

For this task, I connected my laptop to the network and checked the network details using the ipconfig command.

I found the Default Gateway and entered it in the browser. The Jio router login page opened.

Steps

1. Connect the laptop to the network.

2. Open Command Prompt.
3. Type ipconfig.
4. Find the Default Gateway.
5. Enter the gateway address in the browser.
6. The router login page opens.

Figure 1: Jio Router Admin Login Page

TASK 1 – SCREENSHOT / OBSERVATION

The router login page opened after entering the Default Gateway address in the browser.

[INSERT JIO ROUTER LOGIN SCREENSHOT HERE]

Figure 1: Jio Router Admin Login Page

TASK 2 – CHECK IP ADDRESS DETAILS

I opened Command Prompt and used the ipconfig command to check my current network details.

IPv4 Address: 192.168.29.52

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.29.1

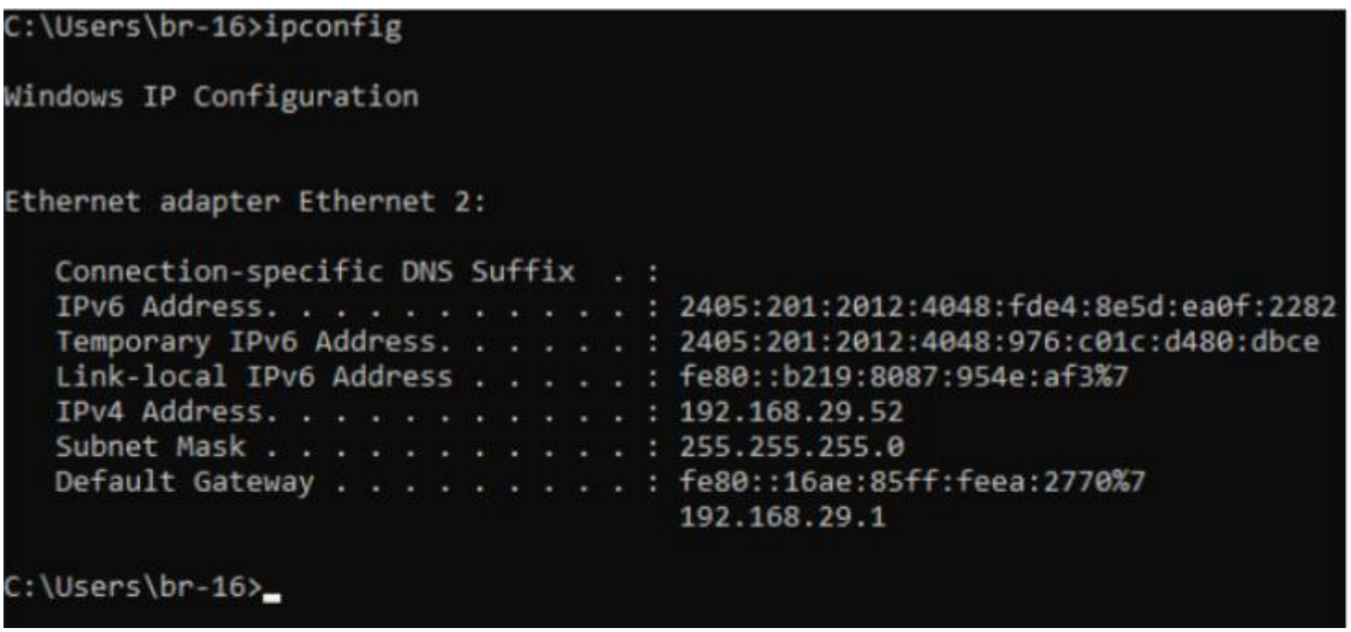

Figure 2: IP Address, Subnet Mask and Default Gateway

TASK 2 – SHORT EXPLANATION

The IP address identifies my laptop on the network. The subnet mask shows the network range. The default gateway is used to communicate with other networks and access the internet.

TASK 3 – CHANGE IP TO STATIC

In this task, I changed the IP setting from automatic (DHCP) to a static IP address in the same network range as the router.

My Current Network Details

- IPv4 Address: 192.168.29.52
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.29.1

Static IP Example

For the practical test, an unused IP address in the same range can be used. Example:

- IP Address: 192.168.29.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.29.1
- Preferred DNS: 8.8.8.8

Note: The static IP should be unused on the network. If the lab network uses a different LAN range, I would use an unused address from that actual range.

TASK 3 – PRACTICAL STEPS

7. Open Network Connections using ncpa.cpl.
8. Right-click the Ethernet/LAN adapter if I am using a LAN cable, then select Properties.
9. Open Internet Protocol Version 4 (TCP/IPv4) Properties.
10. Select Use the following IP address.
11. Enter the static IP, subnet mask and default gateway.
12. Enter the DNS address and click OK.
13. Open a browser and check whether the internet is working.
14. After the test, change the setting back to Obtain an IP address automatically.

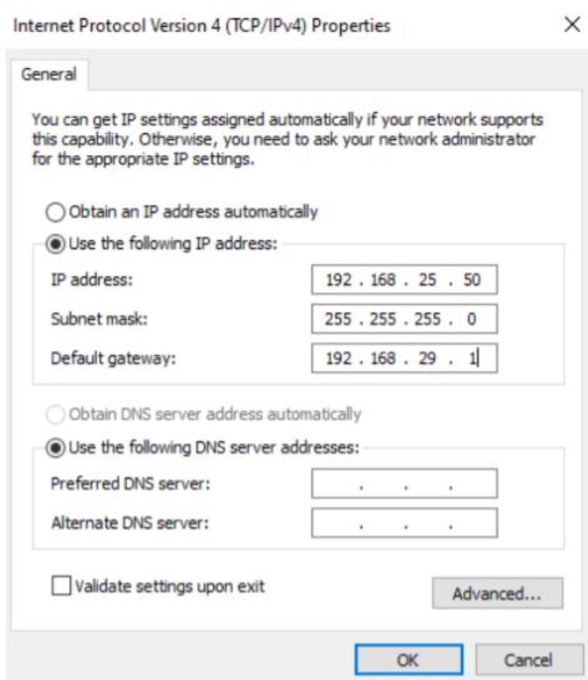

Figure 3: Static IP Address Configuration

Result

This task helped me understand how a static IP is manually configured and how it can be kept in the same network range as the router.

TASK 4 – DHCP RANGE

In this task, I would open the Jio router admin panel and check the DHCP settings. The DHCP section shows the range of IP addresses that the router can automatically assign to connected devices.

Current DHCP Range: 192.168.29.0 – 192.168.29.255

A /24 network has **256 total IP addresses**, from 192.168.29.0 to 192.168.29.255. However, some addresses are reserved for network and broadcast purposes.

What happens if two devices get the same IP?

If two devices are assigned the same IP address, an **IP address conflict** can happen. Because both devices have the same address, communication with the network may not work properly.

Example:

Device A → 192.168.29.20

Device B → 192.168.29.20

Both devices are using the same IP, which can create a **network conflict**.

Screenshot: ***Not included because I do not have access to the router's DHCP settings.***

DAY 3 – PRACTICAL OBSERVATIONS

Network Information

I checked the network configuration of my computer and found the following details:

- IPv4 Address: 192.168.29.52
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.29.1

Router Access

The Default Gateway was used to access the Jio router login page through the browser.

DHCP

DHCP automatically provides IP addresses to devices connected to the network. This makes network configuration easier because the user does not have to enter an IP address manually.

Static IP

A static IP is entered manually and normally stays the same until it is changed. It can be useful when a device needs a fixed network address.

CONCLUSION

In this assignment, I learned the basic concepts of IP addresses, subnet masks, default gateways, DHCP and static IP addresses.

I learned how to use the ipconfig command to check the network details of my computer. I also learned that the Default Gateway can be used to access the router's local login page.

I understood that DHCP gives IP addresses automatically, while a static IP is configured manually. I also learned that using the same IP address on two devices can create an IP conflict.

Overall, this practical helped me understand the basic network configuration of a computer and how a router manages IP addresses for connected devices.

QUICK REVISION

- IP Address = identifies a device on the network
- Subnet Mask = shows the network range
- Default Gateway = connects the local network to other networks
- DHCP = gives IP addresses automatically
- Static IP = manually assigned IP address